ActivePPI: Evaluating Protein-Protein Interaction Networks with Markov Random Fields

Introduction

Proteins interact extensively within cells, forming intricate networks essential for biological functions. Understanding these protein-protein interactions (PPIs) is crucial for deciphering disease mechanisms and therapeutic targets. Despite the wealth of information available in biological databases, most PPIs are recorded without contextual details, making it difficult to determine whether these interactions are active or relevant under biological conditions. To bridge this gap, researchers developed ActivePPI, an innovative computational framework designed to quantify how actively proteins interact within specific cellular environments by leveraging protein mass spectrometry (MS) data.

Motivation Behind ActivePPI

The major challenge in PPI research lies in distinguishing active interactions from general, condition-independent ones listed in databases. Traditional experimental methods, such as yeast two-hybrid screening or co-immunoprecipitation, provide extensive interaction data but typically lack context-specific details. MS-based proteomics, however, offers a quantitative snapshot of protein abundance under specific conditions or diseases, such as cancer or viral infections. Still, integrating these proteomic insights with existing network data has been problematic with conventional analytical tools, as most existing methods do not fully incorporate network structure or quantitative protein data. ActivePPI addresses these limitations by incorporating a robust statistical approach that matches proteomics data directly with known interaction networks.

How ActivePPI Works

ActivePPI uses a statistical modeling approach called Markov Random Fields (MRF). The fundamental assumption in MRF modeling is that the behavior of a protein is influenced primarily by its directly interacting partners. ActivePPI follows a structured approach:

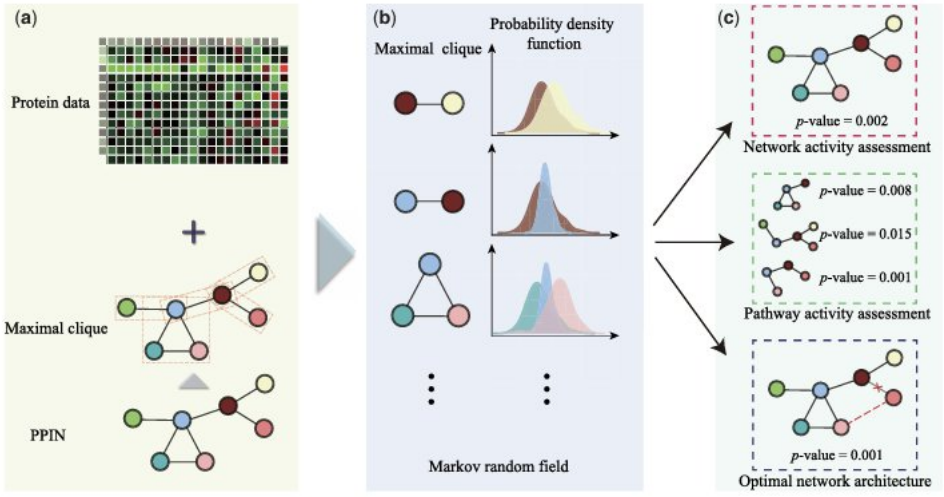

Figure 1. Overview of ActivePPI. (a) Processes protein mass spectrometry data and factorizes PPIN into clusters according to Markov properties. (b) Computes the energy of each maximal clique by estimating the density distribution of proteins. (c) Rewires the network multiple times to calculate the empirical *P*-value for the evaluating PPIN, in the form of network activity, pathway activity, and optimal network architecture.

- **Breaking Down the Interaction Network**:
  The complete network of protein interactions is first decomposed into smaller, tightly connected groups known as maximal cliques. This simplifies analysis by grouping proteins expected to have closely coordinated biological activities.

- **Estimating Clique Activity**:
  Each clique's activity is measured using an energy function. This energy reflects how consistent the protein abundances are within each clique, as measured by MS. A lower energy means the proteins are closely related in their abundance levels, suggesting an active biological interaction within the clique.

- **Statistical Validation**:
  ActivePPI assesses the statistical significance of these network interactions by randomly reshuffling connections multiple times. The original network's measured activity is compared with these randomized networks, allowing calculation of an empirical p-value. Networks with low p-values (below 0.05) are considered significantly active.

Datasets Analyzed

ActivePPI was thoroughly tested using both simulated and real-world datasets to demonstrate its effectiveness and reliability:

- **Breast Cancer (BRCA) Dataset**:
  Real clinical proteomics data was used from the Proteomic Data Commons (PDC). This dataset included protein abundance measurements from 133 breast cancer samples and 18 matched normal tissue samples, covering over 10,491 proteins. Analyzing this dataset enabled ActivePPI to demonstrate its potential for clinical applications by accurately highlighting relevant protein interactions associated with breast cancer.

- **SARS-CoV-2 Dataset**:
  A clinically significant dataset comprising proteomics data from COVID-19 patient autopsies was analyzed. This dataset originally contained MS-derived protein abundance data across multiple organs from 143 infected individuals and 73 healthy controls. After data preprocessing, 7,087 proteins were analyzed. This extensive dataset offered an excellent test case for ActivePPI's applicability to acute viral infections, highlighting protein interactions activated specifically during SARS-CoV-2 infections.

Findings and Observations

ActivePPI's application to these datasets yielded significant insights:

- **Simulated Data Performance**:
  ActivePPI consistently identified known network interactions accurately despite the introduction of random noise. The method remained highly reliable, maintaining statistical significance with empirical p-values typically below 0.01, thus demonstrating robustness in realistic scenarios.

- **Real-world Data Analysis (BRCA and SARS-CoV-2)**:
  In real-world datasets, ActivePPI significantly outperformed existing analytical methods commonly used for protein interaction and pathway analysis, such as GSEA, GSVA, SPIA, and NEAT. It showed superior ability in identifying biologically relevant pathways and protein interactions related specifically to breast cancer progression and COVID-19 severity, indicating its versatility and clinical relevance.

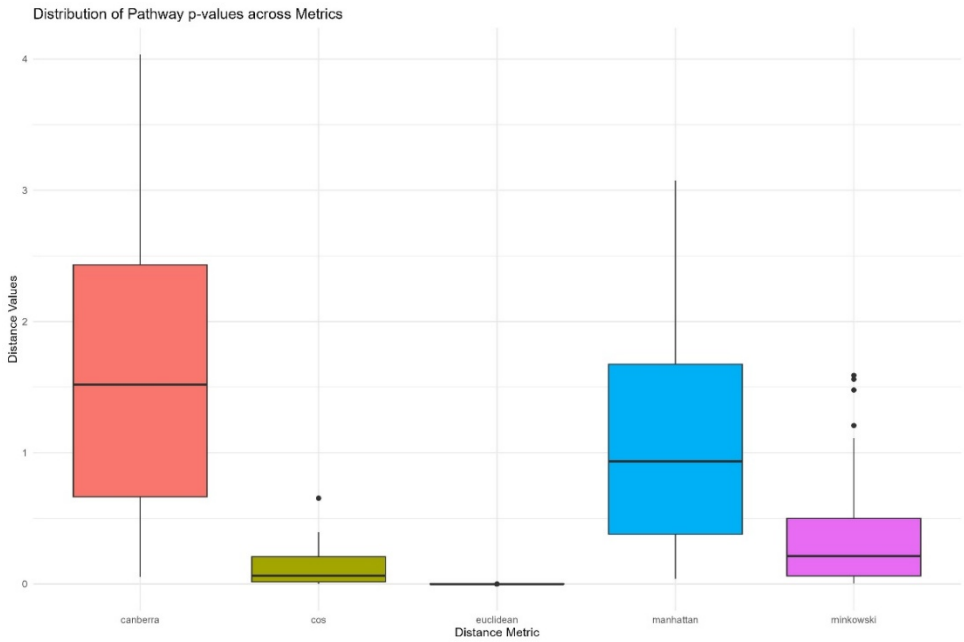

Figure a: Distribution of Pathway p-values across Distance Metrics (BRCA dataset).

- Canberra and Manhattan distances capture large variable differences among pathways. Cosine and Euclidean have very low IQR meaning they fail to capture subtle differences among pathways.

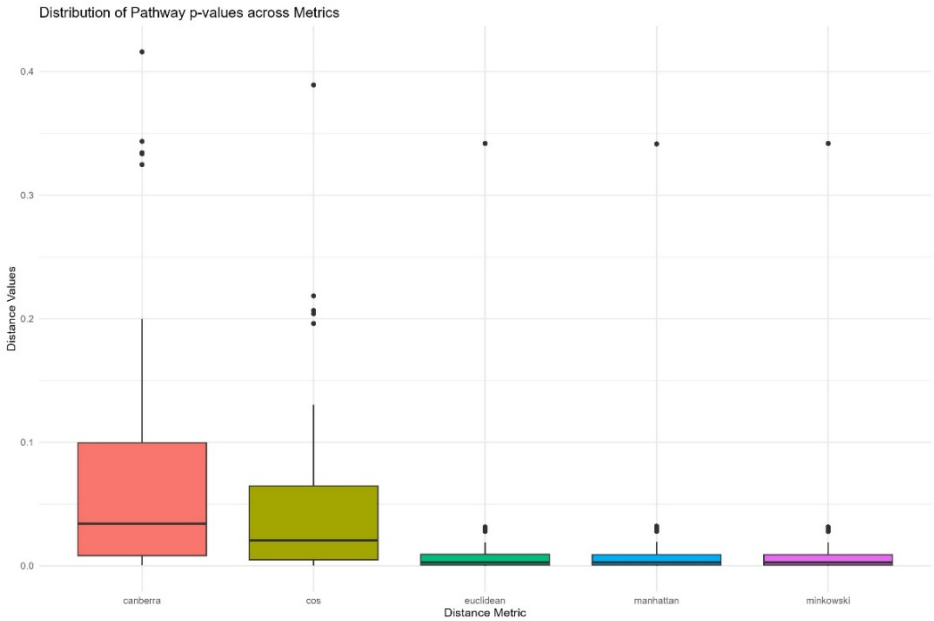

Figure b: Distribution of Pathway p-values across Distance Metrics (SARS-CoV-2 dataset).

- All pathway distances are lower in magnitude as compared to BRCA patients. Most pathways are similar in activity profiles.

- The distribution of pathway p-values across different distance metrics was analyzed for both BRCA and SARS-CoV-2 datasets (Figures a and b). Variations were observed among distance metrics, with Euclidean and Manhattan distances generally showing tighter distributions of lower p-values, indicating better sensitivity in capturing active pathways.

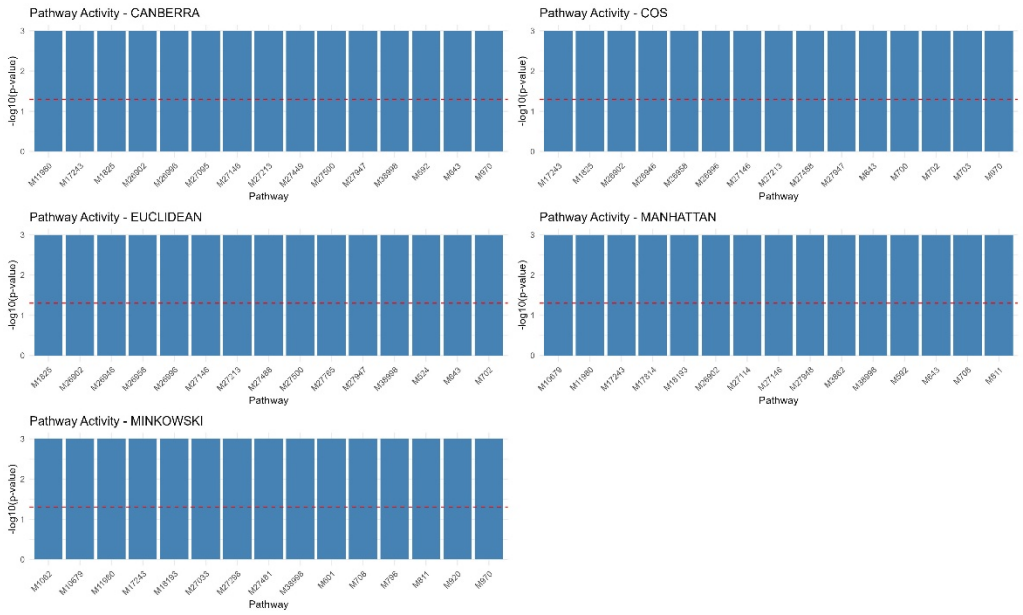

Figure c: Pathway Activity Scores for BRCA dataset across Distance Metrics.

- This plot displaying the pathway significance across the distance metrics in the BRCA dataset shows the pathways identified by ActivePPI with all the pathways showing themselves to be significant as the p-values exceed the threshold within Breast Cancer dataset with identified pathways involved in function such as cell cycle regulation and estrogen dependent gene expression.

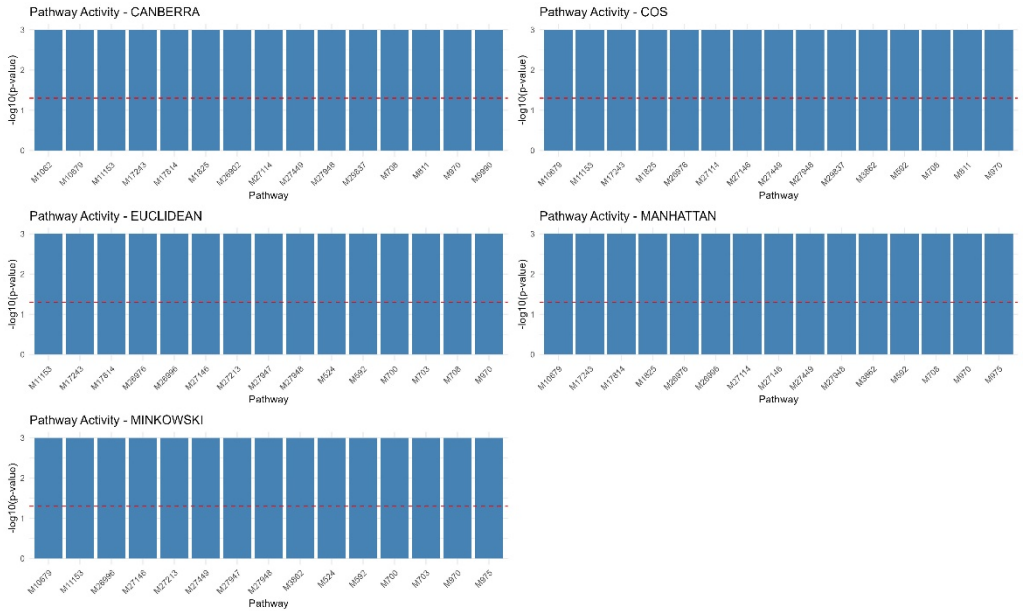

Figure d: Pathway Activity Scores for SARS-CoV-2 dataset across Distance Metrics.

- The plot shows the significances of the pathways in the SARS-COVID dataset, and all the pathways were deemed significant as their $-\log_{10}$ (p-value) had crossed the threshold as some of these are involved in processes such as host entry or mitochondrial immune responses.
- Pathway-specific activity scores were further evaluated, showing that most top pathways consistently exhibited significant activity across different distance metrics, particularly under Euclidean and Manhattan settings (Figures c and d).

Figure e: Top Significant Pathways across Distance Metrics (BRCA dataset).

- The heatmaps support our previous inference from Figure c as some significant pathways can be captured only using certain distance metric.

Figure f: Top Significant Pathways across Distance Metrics (SARS-CoV-2 dataset)

    - This heatmap illustrates the top significant pathways for the SARS-CoV-2 dataset, further confirming that some pathways emerge as significant only under certain distance metrics.

  - The comparison of top pathways across different distance metrics (Figures e and f) highlights how some pathways were consistently ranked among the top performers. This underscores ActivePPI's robustness and reliability in identifying biologically significant pathways irrespective of the specific distance metric used.

- Across both datasets, Euclidean and Manhattan distance metrics consistently provided tighter p-value distributions and more reliable pathway activity detection, indicating their superior sensitivity within ActivePPI's framework.

- **Optimal Network Identification (Case Study: SARS-CoV-2)**:
  In examining the SARS-CoV-2 proteomics dataset, ActivePPI successfully refined the existing network architecture from public databases, highlighting interactions that truly reflected the infection scenario. Notably, it corrected several previously documented but biologically irrelevant interactions. For instance, it identified interactions between proteins RXRA and GCLM as more relevant compared to previously recorded interactions like RXRA and RELA, thus validating its precision in real-world biological contexts.

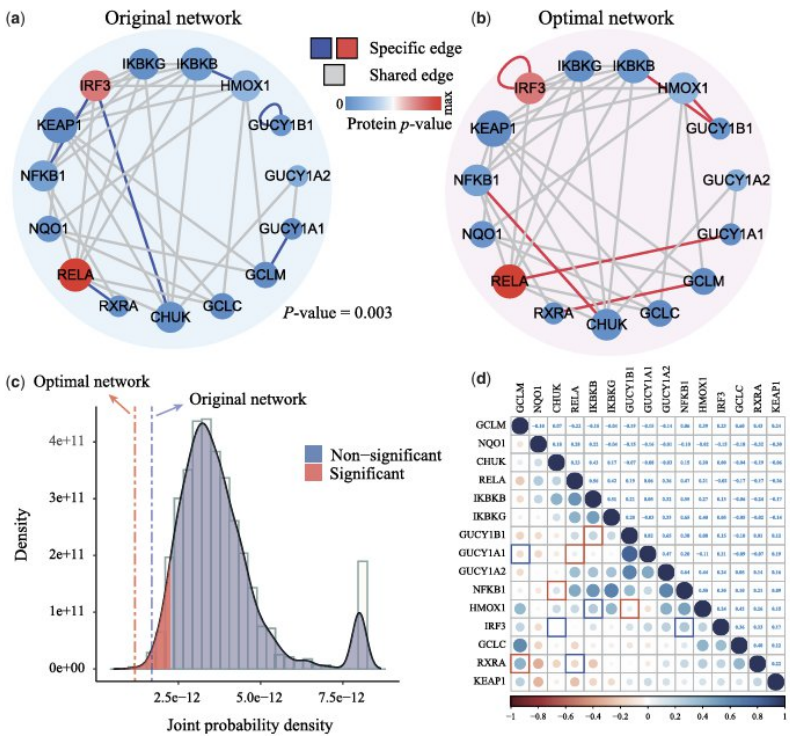

Figure Y: ActivePPI optimization results for the M42560 pathway during SARS-CoV-2 infection. (a) Original network, (b) Optimized network after rewiring, (c) Density distribution showing statistical significance, (d) Protein-protein correlation matrix for validating rewired interactions.

- The empirical P-value calculated for the optimized network was 0.003, strongly supporting the statistical significance of the reorganization identified by ActivePPI under SARS-CoV-2 infection conditions.

Comparative Advantage Over Existing Methods

ActivePPI was compared to several well-established analytical methods categorized into two main groups:

- Set-based methods: CAMERA, GSA, GSEA, GSVA, ORA, SAFE.
- Network-based methods: GANPA, GGEA, LPIA, NEAT, PathNet, SPIA.

In all comparisons, ActivePPI emerged superior, offering a more nuanced and accurate interpretation of proteomic data and network interactions. Its use of MRF to integrate quantitative MS data with PPIs significantly enhanced its accuracy, providing more biologically meaningful insights compared to traditional approaches.

Practical Implications and Future Directions

The practical implications of ActivePPI are considerable, particularly in biomedical research and precision medicine. By accurately determining which protein interactions are active under specific conditions, researchers can better understand disease mechanisms, identify new therapeutic targets, and develop personalized treatment strategies.

Looking ahead, ActivePPI could further evolve by incorporating additional biological data, such as protein structures or genetic sequences. This integration could enhance the accuracy and biological relevance of the method even further, making it a powerful tool for computational biology and clinical research.

Conclusion and Data Accessibility

This project further demonstrates that ActivePPI remains robust across multiple distance metrics and maintains high accuracy even in heterogeneous biological environments such as cancer progression (BRCA dataset) and viral infections (SARS-CoV-2 dataset). Moreover, across all experiments, Euclidean and Manhattan distances consistently provided superior sensitivity in pathway activity detection, further validating the statistical modeling approach used by ActivePPI.

ActivePPI represents a significant advancement in computational biology, offering an effective way to understand and quantify protein interactions under specific biological conditions. It combines the strengths of advanced statistical modeling with high-quality proteomic data, setting a new standard for analyzing PPIs. Moreover, its open-source availability on platforms such as GitHub ensures accessibility and promotes further scientific exploration and validation.

Through these comprehensive analyses and applications, ActivePPI showcases a strong potential to transform how researchers' study and interpret protein interactions in health and disease.

Data Availability:

- ActivePPI source code and datasets: https://github.com/zpliulab/ActivePPI
- BRCA dataset: Proteomic Data Commons, accession number PDC000120 (https://proteomic.datacommons.cancer.gov/pdc)
- COVID-19 proteomics dataset: Nie et al., 2021, available via supplemental data at *Cell*: https://doi.org/10.1016/j.cell.2021.01.004

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These references provide foundational background and detailed supporting evidence for all aspects discussed in the report.